$\frac{1}{24} + \frac{1}{8}$ $\quad 1 + \frac{3}{5}$ $\quad \frac{8}{5}$ $\quad \frac{1}{3} + \frac{1}{5}\left(\frac{1}{3}\right)$ $\quad \frac{1}{3} + \frac{1}{12}$

EE595H: Quiz

Total Questions: 2, Total Marks: 15, Time: 60 Mins

1. Consider a Markov chain $\{X_n, n \geq 0\}$ with states 0, 1, 2. Suppose it has the transition probability matrix

$$\begin{bmatrix} \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ 0 & \frac{1}{3} & \frac{2}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}$$

If $P(X_0 = 0) = P(X_0 = 1) = 1/4$, find $E[X_3]$. $\boxed{7}$

2. Consider a Markov chain with the following transition probability matrix:

$$\begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{4} & 0 & \frac{3}{4} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}.$$

(a) Draw the state transition diagram. $\boxed{2}$

(b) Find the stationary distribution. $\boxed{6}$

$\boxed{\text{Total: 8}}$